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How to use NEON SIMD to accelerate c++ scripts

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ProbablyHuman
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Posts: 42

Joined: Fri Jun 12, 2020 10:19 pm

How to use NEON SIMD to accelerate c++ scripts

Mon Aug 10, 2020 3:12 am

Hi, I have a RPI-3b+ running ubuntu 16.04, and I was using a library called 'usb_cam' to grab the feed of a camera. Unfortunately, there was some latency with the displayed video and real-life. I thought that the best approach would be optimization, so i made a fork of the repository (https://github.com/ai-are-better-than-humans/usb_cam) and changed the code a bit, yet when I ran 'htop' i still had one of my cores maxed out (<https://imgur.com/gallery/d9xrqxG?s=sms> (<https://imgur.com/gallery/d9xrqxG?s=sms>)). I asked around, and the fault was not that of the code, but just the sheer amount of pixels the 650x480 camera had, and its lack of onboard compression (Im using an ov7251 with a Y10/Y10P pixel format).

They suggested using something called parallelization/auto-vectorization. A bit more research and I found out that the rpi-3b+ supported VFPv4 and NEON as their FPU's. While I do know that these will help me speed up the processes, I still can't figure out how to utilize them, no matter how much i look. So my question is: how do I use neon's simd capabilities to speed up my script and take less of a load on the cpu? Please do keep the instructions low level, i use these terms not because i am experienced with them but because they are related to the subject at hand... lol. =)



jahboater

Posts: 9199

Joined: Wed Feb 04, 2015 6:38 pm

Re: How to use NEON SIMD to accelerate c++ scripts

Mon Aug 10, 2020 7:02 am

ProbablyHuman (./memberlist.php?mode=viewprofile&u=338704) wrote: (./viewtopic.php?p=1710227#p1710227)
Mon Aug 10, 2020 3:12 am

They suggested using something called parallelization/auto-vectorization. A bit more research and I found out that the rpi-3b+ supported VFPv4 and NEON as their FPU's. While I do know that these will help me speed up the processes, I still can't figure out how to utilize them, no matter how much i look. So my question is: how do I use neon's simd capabilities to speed up my script and take less of a load on the cpu? Please do keep the instructions low level, i use these terms not because i am experienced with them but because they are related to the subject at hand... lol. =)

Firstly C++ is not a scripting language. The program source code is compiled to an executable (much faster).

This is non-trivial. Here are some possible things to look at:

You could try the "-ffree-vectorize" compiler option (and maybe others), to hopefully get the compiler to automatically vectorize things for you.

You can try using the NEON intrinsic's, documented here:

<https://gcc.gnu.org/onlinedocs/gcc-4.6.3/nsics.html> (<https://gcc.gnu.org/onlinedocs/gcc-4.6.4/gcc/ARM-NEON-Intrinsics.html>)

You can try inline assembler, documented here:

[https://gcc.gnu.org/onlinedocs/gcc-10.2 ... age-with-C](https://gcc.gnu.org/onlinedocs/gcc-10.2...age-with-C) (<https://gcc.gnu.org/onlinedocs/gcc-10.2.0/gcc/Using-Assembly-Language-with-C.html#Using-Assembly-Language-with-C>)

You should use the latest version of the compiler, which is GCC 10.2 at the time of writing:

<https://gcc.gnu.org/> (<https://gcc.gnu.org/>)

Make sure the right compiler options are in effect, say: `-mcpu=native`

Easy option:

Consider getting a Pi4, it is much faster (and it may be trivially overclocked to 2GHz or more).

The Pi4 also comes with 2GB of memory minimum, going up to 8GB.

The Pi4 runs a 64-bit OS if you like (which affects the way NEON works - for the better).

ttapa

Posts: 199

Joined: Mon Apr 06, 2020 2:52 pm

Re: How to use NEON SIMD to accelerate c++ scripts

Mon Aug 10, 2020 11:56 am

You can find my notes on the topic here: [Raspberry-Pi/NEON](https://ttapa.github.io/Pages/Raspberry-Pi/NEON/index.html) (<https://ttapa.github.io/Pages/Raspberry-Pi/NEON/index.html>). It contains some very useful links to the ARM documentation, and some example code.

I've also experimented with NEON for doing alpha overlays (overlaying an image with transparency over another image), you can find the code here: <https://github.com/ttapa/ARM-NEON-Compositor> (<https://github.com/ttapa/ARM-NEON-Compositor>).

Many open-source libraries that handle graphics will have ARM NEON implementations for some of the core algorithms, so you can use those as an example as well. Some are implemented in assembly, some use intrinsics. Sadly, they often aren't documented that clearly.

This is not something that you learn overnight, there's a lot you need to know about how SIMD works, what intrinsics/instructions exist, and how you can transform your original code to something that can easily be parallelized using SIMD.

Before trying SIMD intrinsics, you can try using OpenMP to parallelize a loop over multiple threads/cores, or you can tell GCC that there are no dependencies in your loop using `#pragma ivdep` (<https://gcc.gnu.org/onlinedocs/gcc/Loop-Specific-Pragmas.html>), so the optimizer can do its job more effectively, using SIMD instructions if possible (and if you enabled vectorization using optimization flags).

The unfortunate truth is that most pieces of code are not optimized using SIMD by the compiler, and a hand-coded solution is still much faster. GCC 10 did bring some improvements, but if you want the compiler to optimize it for you, you often need to structure your code in such a way that it can easily spot SIMD opportunities, and you have to make sure that there are no (hidden) dependencies between data.

You can use the `-fopt-info-vec-all` GCC option to see where the compiler was able to vectorize loops or why it failed.

Pieter





jahboater

Posts: 9199

Joined: Wed Feb 04, 2015 6:38 pm

Re: How to use NEON SIMD to accelerate c++ scripts

Mon Aug 10, 2020 12:20 pm

tttapa (./memberlist.php?mode=viewprofile&u=330701) wrote: (./viewtopic.php?p=1710376#p1710376)

Mon Aug 10, 2020 11:56 am

You can find my notes on the topic here: [Raspberry-Pi/NEON \(https://tttapa.github.io/Pages/Raspberry-Pi/NEON/index.html\)](https://tttapa.github.io/Pages/Raspberry-Pi/NEON/index.html). It contains some very useful links to the ARM documentation, and some example code.

That's a great resource, thanks for posting it.

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